Matteo Martini

Nationality: Italian **Date of birth:** 21/11/1994 **Gender:** Male

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EDUCATION AND TRAINING

Ph.D in Computer Science

University of Genoa [01/11/2023 – Current]

City: Genoa | **Country:** Italy | **Field(s) of study:** Virtual Reality ; Extended Reality ; Human-Computer Interaction | **Level in EQF:** EQF level 8

Master's Degree in Computer Science

Università di Genova [01/09/2020 – 29/03/2023]

City: Genova | **Country:** Italy | **Field(s) of study:** Virtual/Mixed Reality, Deep Learning, Stereopsis, Computer Vision | **Final grade:** 110 e Lode | **Level in EQF:** EQF level 7 | **Number of credits:** 120 | **Thesis:** Dynamic obstacle avoidance in Mixed Reality - An approach based on deep learning and stereopsis

The thesis focused on the development of a system that combines the ZED camera and YOLO (You Only Look Once) algorithm to recognize real-world objects and seamlessly integrate them into virtual scenes with consistent position and size. The software has been designed initially for non-intrusively informing users of a VR application about obstacles in the physical room. However, it has been tested and proven adaptable for various purposes, including the manipulation of real objects connected to their virtual counterparts to leverage passive haptic feedback.

Advisors: Manuela Chessa, Fabio Solari
Examiner: Enrico Puppo

Bachelor's Degree in Computer Science

University of Genoa [01/09/2017 – 29/10/2020]

City: Genoa | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies | **Final grade:** 110 cum laude | **Level in EQF:** EQF level 6 | **Number of credits:** 180 | **Thesis:** Pi-Calculus: A Formalism For Distributed Computations

Analysis of Pi-Calculus, a formal language for describing distributed computations. Study of its formalization by observing the external vision of a system, modeled by reduction and structural congruence, the internal vision, modeled by the labeled transition, and the Harmony lemma which establishes the coherence between the two.

Supervisors: Francesco Dagnino, Elena Zucca

High School Diploma - Accountant and Programmer

IIS G. Ruffini [01/09/2008 – 01/06/2013]

City: Imperia | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies | **Final grade:** 83 | **Level in EQF:** EQF level 5

WORK EXPERIENCE

University of Genoa – Genoa, Italy

City: Genoa | **Country:** Italy

Research fellow

[01/08/2023 – 31/10/2023]

Research grant: Interaction systems for cognitive exergames based on VR environments.
Research supervisor: Prof. Manuela Chessa

🏢 ***University of Cagliari*** – Cagliari, Italy

City: Cagliari | **Country:** Italy

Research fellow

[01/04/2023 – 30/06/2023]

Research Grant: Development of Immersive and Non-immersive Virtual Reality Applications Integrating Motion Analysis Sensors
Research Supervisors: Dr. Giulia Sedda, Prof. Danilo Pani

The objective of this research project was to develop an immersive and non-immersive Virtual Reality application that integrate motion analysis sensors. Specifically, the focus was on designing and implementing a serious game using Unity for the upper limb rehabilitation of patients affected by Rett syndrome. The motion analysis was performed using the ZED camera.

🏢 ***University of Genoa*** – Genoa, Italy

City: Genoa | **Country:** Italy

Teacher for Unige Senior

[01/2023 – 05/2023]

Advanced level Computer Science course. Topics covered:

- Smartphones and tablets - hardware, software, structure, connectivity, and settings
- Cloud - remote storage, applications, services
- Digital life - digital identity (SPID, CIE), digital signature, PEC (Certified Electronic Mail), relations with public administration, personal blog
- Introduction to new technologies based on AI

🏢 ***University of Genoa*** – Genoa, Italy

City: Genoa | **Country:** Italy

University didactic tutor

[10/2022 – 05/2023]

Support for first-year students in the Bachelor's degree program in Computer Science for the courses Introduction to Programming, Algorithms and Data Structures, Computer Architecture. Classroom lectures and assistance during practical labs.

🏢 ***Aqua DB s.r.l.*** – Imperia, Italy

City: Imperia | **Country:** Italy

Collaborator

[2015 – 2016]

Software remote installation and assistance, development of customizations requested by clients.

🏢 – Imperia, Genoa, Italy

City: Imperia, Genoa | **Country:** Italy

Photographer and video maker

[2015 – Current]

Creation of photos and videos of various genres, interviews, documentaries, music videos, short films.

PUBLICATIONS

[2025]
Immersive versus Non-Immersive Virtual Reality Environments: Comparing Different Visualization Modalities in a Cognitive-Motor Dual-Task In fields like cognitive and physical rehabilitation, adopting immersive visualization devices can be unfeasible. In these cases, the main challenge is to develop Virtual Reality (VR) scenarios that still provide a strong sense of presence, usability, and user agency, even without full immersion. This paper explores a cognitive motor dual-task in VR, consisting in counting and reaching, comparing three non-immersive visualization methods on a 2D screen (tracked perspective camera, fixed perspective camera, fixed orthographic camera) with the immersive experience provided by a head-mounted display. The comparison focused on factors like sense of presence, usability, cognitive load, and task accuracy. Results show, as expected, that immersive VR provides a higher sense of presence and better usability with respect to the non-immersive visualization methods. Unexpectedly, the implemented 2D visualization based on a tracked perspective camera seems not to be

the best approximation of immersive VR. Finally, the two fixed camera conditions showed no significant differences in performance based on the type of projection.

Authors: Marianna Pizzo, Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name**: VISIGRAPP | **Publisher**: Science and Technology Publications

[2025]
Cooperation in Virtual Reality: Exploring Environmental Decision-Making through a Real-Effort Threshold Public Goods Game We describe our contribution to the IEEE 3DUI Contest for IEEE VR 2025. Specifically, we present a collaborative Virtual Environment (VE) to be used as an innovative method to conduct experimental research in behavioral economics studies. Indeed, in such a field, analyzing collaborative behaviors is crucial, and collaborative VEs might be exploited to better control the variables of interest while preserving a high sense of realism and presence, to spread collaboration across wider geographical areas, and to maintain anonymity while preserving the sense of being with peers in-side environments. The developed system aims to be used as a general framework to perform a wide range of studies in the field, here explicitly focusing on a real-effort recycling task, where subjects decide whether to prioritize individual gains or contribute to a collective effort to meet a common recycling objective.

Authors: Manuela Chessa, Michela Chessa, Lorenzo Gerini, Matteo Martini, Kaloyana Naneva, Marianna Pizzo, Fabio Solari, Paolo Zeppini | **Journal Name**: IEEE VR (Abstracts and Workshops) | **Publisher**: IEEE

[2025]
Virtual Reality as a Tool for Upper Limb Rehabilitation in Rett Syndrome: Reducing Stereotypies and Improving Motor Skills Rett Syndrome (RTT) is a rare neurodevelopmental disorder that causes the loss of motor, communicative, and cognitive skills. While no cure exists, rehabilitation plays a crucial role in improving quality of life. Virtual Reality (VR) has shown promise in enhancing motor function and reducing stereotypic behaviors in RTT. This study aims to assess the impact of VR training on upper limb motor skills in RTT patients, focusing on reaching and hand-opening tasks, as well as examining its role in motivation and engagement during rehabilitation.Twenty RTT patients (aged 5–33) were randomly assigned to an experimental group (VR training) and a control group (standard rehabilitation). Pre- and post-tests evaluated motor skills and motivation in both VR and real-world contexts. The VR training involved 40 sessions over 8 weeks, focusing on fine motor tasks. Non-parametric statistical methods were used to analyze the data.Results indicated significant improvements in the experimental group for motor parameters, including reduced stereotypy intensity and frequency, faster response times, and increased correct performance. These improvements were consistent across VR and ecological conditions. Moreover, attention time increased, while the number of aids required decreased, highlighting enhanced engagement and independence. However, motivation levels remained stable throughout the sessions.This study demonstrates the potential of VR as a tool for RTT rehabilitation, addressing both motor and engagement challenges. Future research should explore the customization of VR environments to maximize the generalization of skills and sustain motivation over extended training periods.

Authors: Rosa Angela Fabio, Martina Semino, Michela Perina, Matteo Martini, Emanuela Riccio, Giulia Pili, Danilo Pani, Manuela Chessa | **Journal Name**: Pediatric Reports | **Publisher**: MDPI

[2025]
Deep Learning-Based Classification of Social Anxiety Disorder Using Continuous Self-Reported Anxiety in Virtual Reality S ocial Anxiety Disorder (SAD) is a mental disorder characterized by excessive fear and avoidance of social situations. Traditional assessment methods rely on retrospective self-reports, which may not fully capture moment-to-moment variations in perceived anxiety. To address this, we designed a novel Virtual Reality (VR) scenario to simulate a real-life social situation, specifically a waiting room that gradually fills with virtual characters. A continuous measure of self-reported anxiety was collected via joystick throughout the VR experience, allowing for real-time monitoring of subjective social anxiety. A one-dimensional convolutional neural network (1D-CNN) was trained to classify individuals with SAD based on their reported anxiety trajectories. The model was evaluated using a Leave-One-Subject-Out (LOSO) cross-validation strategy, achieving an F1-score of 0.82, recall of 0.89, and precision of 0.77, demonstrating strong classification performance. These findings suggest that self-reported anxiety alone is a viable signal for distinguishing individuals with SAD, paving the way for more accessible, sensor-free assessment tools in virtual environments. Future work will explore advanced feature extraction from the anxiety signal, integrate physiological markers, and investigate adaptive VR scenarios that dynamically respond to user-reported distress.

Authors: Marco Pardini, Sergio Frumento, Matteo Martini, Gianluca Rho, Valerio Vatteroni, Krishant Tharun, Martina Alaimo, Federico A Galatolo, Martina De Marinis, Enzo Pasquale Scilingo, Danilo Menicucci, Mario GCA Cimino, Manuela Chessa, Alberto Greco | **Journal Name**: IEEE MeMeA | **Publisher**: IEEE

[2025]
A Virtual Room for Rett Syndrome: Design and Implementation of an Exergame for Upper-Limb Rehabilitation Rett syndrome (RTT) is a rare neurodevelopmental disorder that severely impairs motor and cognitive functions, posing significant challenges for rehabilitation. This paper presents Virtual Room 2.0, a non-immersive Virtual Reality (VR) exergame designed to promote upper limb rehabilitation in RTT patients through engaging, interactive exercises. The system is based on marker-less

body tracking obtained via a stereo camera, which allows its use in non-laboratory environments, such as homes and day centers. A 40-session, 10-week training protocol was conducted to assess the system's effectiveness, involving 10 RTT patients of different ages and clinical stages. Motion data were collected and analyzed to assess improvements in reaching movements. Results indicate a positive trend in upper limb function as well as increased motivation and engagement, confirming the potential of non-immersive VR-based rehabilitation for RTT patients.

Authors: Matteo Martini, Giulia Pili, Martina Semino, Michela Perina, Emanuela Riccio, Rosa Angela Fabio, Danilo Pani, Manuela Chessa | **Journal Name**: IEEE ISMAR (Adjunct) | **Publisher**: IEEE

[2024]
IMMERSE: IMMersive Environment for Representing Self-Avatar Easily Recently, the importance of having a representation of the users inside Virtual Reality (VR) by using avatars, which replicate the movements of real humans, has grown in several fields of application. Nowadays, developers can use many off-the-shelf devices for visualizations and tracking. Game engines like Unity3D and Unreal allow to easily integrate packages to manage head-mounted displays and other tracking devices. However, it is still difficult to combine different solutions and achieve a full representation of the human body, e.g., combining full-body and hand tracking or switching among different tracking modalities in an easy way. This paper describes IMMERSE, an open-source framework based on Unity3D, the XR Interaction toolkit, and an inverse kinematics solver. IMMERSE allows developers to insert a VR avatar in a virtual environment, animate it through different sources of 6DOF tracking measurements, which are also interchangeable during the simulation, and finally record the movements to animate other avatars or input to motion analysis techniques.

Authors: Eros Viola, Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name**: IEEE GEM | **Publisher**: IEEE

[2024]
Designing an Immersive Virtual Reality Scenario for Social Anxiety Elicitation and Modeling: A Preliminary Evaluation We describe an immersive Virtual Environment (VE) designed and developed to record physiological and behavioral data to identify a computational model for real-time subjective anxiety estimation. The VE represents a waiting room with several chairs. Virtual Reality (VR) avatars come into the room and sit around the subject, sitting each in one of the available chairs, following two different strategies. Different forms of interaction between the avatars and the subject have been implemented. This paper presents preliminary tests of the VE conducted with 24 healthy volunteers self-reporting possible symptoms of Social Anxiety Disorder, who were invited to continuously report their subjective level of discomfort through the VR controller. Once integrated with the answers given to questionnaires aimed at investigating anxiety states and immersiveness in the VE, these preliminary results demonstrated the effectiveness of the proposed scenario as a novel and useful tool for robustly eliciting and assessing states of social anxiety in a realistic but nevertheless controllable and acceptable way.

Authors: Matteo Martini, Eros Viola, Francesco Bossi, Sergio Frumento, Alessio Iannizzotto, Sara Said, Alejandro Luis Callara, Fabio Solari, Enzo Pasquale Scilingo, Alberto Greco, Danilo Menicucci, Manuela Chessa | **Journal Name**: IEEE MetroXRAINE | **Publisher**: IEEE

[2024]
Extended Reality and Artificial Intelligence for Exergaming: Opportunities and Open Challenges for Rehabilitation and Cognitive Training The increased interest in the use of Virtual, Augmented, and Mixed Reality technologies, also referred to by the wider term eXtended Reality (XR), to create effective and innovative serious games and exergames in the healthcare domain opens several questions and points out open and still unsolved issues. In this paper, we will describe the main available software and hardware technologies and discuss the opportunities and open challenges in rehabilitation. We will also discuss the need to fill the existing gap between the advancement of techniques in the XR domain and their effective use in healthcare. Specifically, we will describe some use cases recently developed and included in clinical trials.

Authors: Manuela Chessa, Lorenzo Gerini, Razeen Hussain, Matteo Martini, Marianna Pizzo, Fabio Solari, Eros Viola | **Journal Name**: IEEE RTSI | **Publisher**: IEEE

[2023]
Obstacle Avoidance and Interaction in Extended Reality: An Approach Based on 3D Object Detection In recent years, the compactness and capabilities of modern head-mounted displays have rekindled interest in Virtual Reality. There are now various options available, from discrete to integrated all-in-one solutions that no longer require an expensive workstation for graphical rendering. This has made it possible for a larger audience to own a personal VR device.Despite manufacturers' safety recommendations, the home use of these devices can be dangerous. The currently available devices are equipped with basic safety systems that require de��ning a safe zone and just worn users whenever they are going to move outside of it. However, these systems are not capable of detecting obstacles in a realistic home environment. For example, living rooms or bedrooms may be cluttered with furniture and other objects that are not easily removable or that users will not remove, making it easy for them to hit something or trip over them.To address this problem, we propose an obstacle avoidance system that combines state-of-the-art object detection with depth perception capabilities granted by stereopsis in RGB-D cameras. Our system can detect

speciﬁc classes of objects, estimate their position and extent, and create a virtual counterpart for each one of them in an immersive virtual scene. This will help users visualize obstacles in their environment and avoid collisions.

Authors: Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name**: International Conference on Image Analysis and Processing | **Publisher**: Springer Nature Switzerland

SCHOOLS AND COURSES

[17/07/2023 – 21/07/2023]

European Agent Systems Summer School (EASSS)

I attended the 23th edition of EASSS held in the Czech Technical University in Pragu, which aims to provide exchange of knowledge among individuals and groups interested in various aspects of autonomous systems. This dissemination is provided by formal state-of-the-art courses conducted by leading experts in the field and by informal meetings during the event. The school attracts both beginner and experienced researchers, encouraging cooperation between representatives of many branches of Multi-Agent Systems research community.

[04/03/2024 – 08/03/2024]

Social XR Spring School (CWI, DIS)

I attended the Social XR spring school organized by the DIS lab at CWI in Amsterdam, which aims to explore this new medium for communication and collaboration through an interdisciplinary approach. The program covered several key areas, including:

- 3D capturing and modeling of realistic avatars and their behavior in the virtual world.
- Coding, transmission, and distribution of volumetric video content, along with optimization techniques to improve the quality of experience (QoE).
- Ethics by design, with an emphasis on developing responsible Social XR experiences.
- Rendering and interaction, which involved creating new immersive and multi-sensory experiences.
- Human factors and evaluation, focusing on the assessment of experiences and the modeling and prediction of QoE.

OTHER EXPERIENCES

Open days for high school students

Introductory sessions for visiting high school students who are considering enrolling in the Bachelor of Science in Computer Science program. We explained and let them experience the differences between Virtual Reality and Mixed Reality and showed them past and ongoing projects that the Perception and Interaction Lab is working on.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

Inglese

LISTENING C1 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user